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2.4.5 Funaria

Class – Bryopsida
Order- Funariales
Family – Funariaceae
Genus - *Funaria*

Funaria is commonly called 'cord moss'. It is distributed throughout the world. *Funaria hygrometrica* is the common species. It grows in close tufts on rocks, trunks of trees, damp walls and damp soils. They help in the process of soil formation (Pedogenesis).

External features

The plant body is a gametophyte . It is small, 1 to 3 cm high and consists of slender erect radial stem covered with small, simple leaf like structures arranged in a spiral manner. The gametophyte is attached to the substratum by means of multicellular rhizoids. They are characterized by the presence of oblique septa. The leaves are simple, sessile ovate and have broad membranous base and pointed apex.

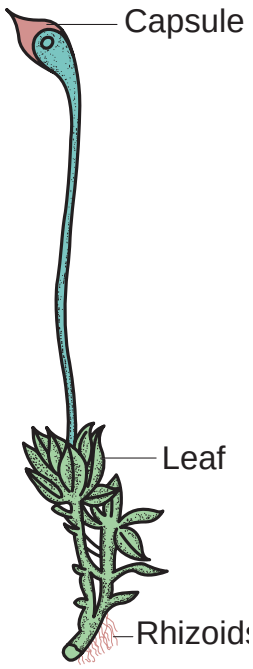


Figure 2.18: *Funaria* Habit

Internal structure

T.S. of axis

The T.S. of axis shows the presence of epidermis, cortex and central cylinder. The epidermis is the outermost layer and contain chloroplast bearing cells. The cortex is made up of parenchymatous tissue. The cells of the young axis bear chloroplasts. In mature stems the outermost cells become reddish brown colour and become thick walled. Small leaf traces are also noticed. The central cylinder is made up of long, narrow, thin walled, colourless cells which lack protoplasts. They help in the conduction of water and minerals.

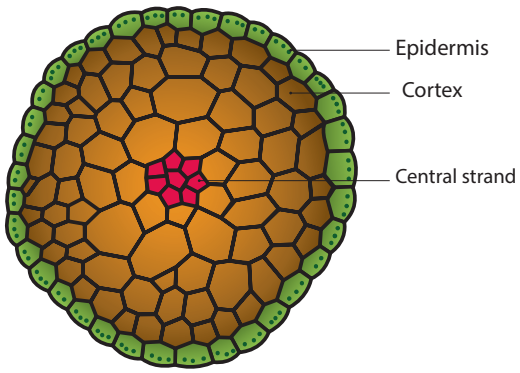


Figure 2.19: T.S. of axis

T.S. of leaf

A well defined midrib is present. It consists of several layers of cells but the lateral 'wing' or lamina is made up of single layer of thin walled cells which are rich in chloroplasts. Midrib contains small strands of slightly thickened narrow cells which help in conduction.

Reproduction

Funaria reproduces by vegetative and sexual methods.

Vegetative reproduction

Vegetative reproduction takes place by the following methods (Figure 2.20):

1. Fragmentation of primary protonema,
2. Formation of secondary protonema from any part of the gametophyte
3. Formation of gemmae on terminal cells of the protonema.